Enhanced Portfolio Optimization: Integrating ONS with Volatility Thresholds for Adaptive Asset Allocation

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Motivation

- Portfolio selection technique is important to balance risk exposure and expected return
- Classic Markowitz mean-variance selection model assumes a given stock's return and risk remain unchanged based on historical data
  - ➢ Huge amount of data sample is required
- Online Newton Step (ONS) is one of the five types of online learning algorithm
  - ➢ We aim to improve the performance of a portfolio composed using ONS by considering side information

Model Formulation

- Number of available stocks is $n, n \in N^+$
- Relative price change vector $\boldsymbol{r_t} \in R^n, r_{t,i} = \frac{S_{t,i} - S_{t-1,i}}{S_{t-1,i}}$
  - ➢ $S_{t,i}$ is the closing price of stock i on day t
- Daily portfolio is $\boldsymbol{p_t}$ and logarithmic wealth growth over time is $\sum_{\tau=0}^{K} \log(\boldsymbol{p_\tau} \cdot r_\tau)$
  - ➢ An investor seeks to $\max_{\boldsymbol{p_\tau} \in S_n} f = \sum_{\tau=0}^{K} \log(\boldsymbol{p_\tau} \cdot r_\tau)$
- Online Newton Step uses 2nd order Newton's step to iteratively find roots to a 1st order derivative of $f(x)$, s.t,

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$$

- In higher dimensions, gradient descent matrix (b_t) replaces $f'(x_n)$ and Hessian matrix (A_t) replaces $f''(x_n)$, s.t,

$$x_{n+1} = x_n - [\nabla^2 f(x_n)]^{-1} \nabla f(x_n)$$

- It is equivalent to solve the following optimization problem

$$\min_{x \in S_n} \|\boldsymbol{y_t} - \boldsymbol{x}\|^2$$

$$\text{subject to } -x_{t,i} \leq 0\ , i \in n, t \in \{1,2, \dots, T\}$$

$$\sum_i x_{t,i} = 1\ , t \in \{1,2, \dots, T\}$$

where $y_t = A_{t-1}^{-1} b_{t-1}$ and $p_t = \prod_{S_n}^{A_{t-1}} y_t$

- We added additional constraint to the above problem
  - ➢ Volatility: check if $stock_i$ exceeds volatility threshold V in the past K days, up to $t-1$ day
    - ➢ If true: set $x_{i,t} = 0$ and standardize remaining weights
    - ➢ If false: maintain current weight in the portfolio on day t

ONS Algorithm

Algorithm 1 Online Newton Step

Input: convex set S_n, T, $x_1 \in S_n \subseteq R^n$, parameters $\delta, \beta > 0$, $A_0 = I_n$
$t = 0, p_0 = (1/n)\mathbf{1}$
for $t = 1$ to T **do**
  Play p_t and observe cost $f_t(p_t)$.
  Update: $A_t = A_{t-1} + \nabla f_t \nabla f_t^T$
  Update: $b_t = b_{t-1} + (1 + 1/\beta) * \nabla f_t$
  Newton step and generalized projection:
  $y_{t+1} = x_t - A_t^{-1} b_t$
  $p_{t+1} = \Pi_K^{A_t}(y_{t+1})$
end for

Algorithm 2 Online Newton Step + Volatility Filter

Input: convex set S_n, T, $x_1 \in S_n \subseteq R^n$, parameters $\delta, \beta > 0$, $A_0 = I_n$
$t = 0, p_0 = (1/n)\mathbf{1}$
for $t = 1$ to T **do**
  Play p_t and observe cost $f_t(p_t)$.
  Update: $A_t = A_{t-1} + \nabla f_t \nabla f_t^T$
  Update: $b_t = b_{t-1} + (1 + 1/\beta) * \nabla f_t$
  Newton step and generalized projection:
  $y_{t+1} = x_t - \delta A_t^{-1} * b_t$
  if $Stock_i$ volatility in past K periods exceeds threshold V **then**
    $y_{t+1,i} = 0$ and $y_{t+1,j} = y_{t+1,j} / \sum y_{t+1,j}$
  end if
  $p_{t+1} = \Pi_{S_n}^{A_t}(y_{t+1})$
end for

- Run time is $O(n^3)$ for both algorithm

Results/Discussion

- Benchmarked against QQQ and available stocks from QQQ holding list (~100)
- Tuned parameters using data from 2020/3/16 to 2023/3/15
  - ➢ Parameters can be adjusted to fit different risk appetite investors
- ONS with side information achieved higher return than ONS itself at a lower risk level
- Volatility threshold can be adjusted as well
- Richer training data provides better return using ONS with volatility control as side information
- Slow run time prevents such an algorithm to execute exactly described in literature
  - ➢ Requires investor to execute before market closes
  - ➢ Prices used for calculation may change during the process of executing the algorithm

2023/3/16 - 2024/3/15	ONS	ONS+Vt	Universal	QQQ
Cumulative Return	0.8901 (0.2445)	0.9901 (0.2413)	0.3399 (0.1501)	0.4464 (0.1614)
Sharpe Ratio	3.437	3.896	2.149	2.456

Note: $\delta = 1, \beta = 0.01. K = 10$ days, $V = 0.05$. Standard deviation in parenthesis. Annualized risk free rate 0.05

Table 1: Result Comparison

2023/3/16 - 2024/3/15	K=3, V=0.01	K=5, V= 0.03	K=10, V=0.05	K=15, V=0.07	K=20, V=0.1
Cumulative Return	0.5862 (0.1869)	0.3872 (0.1744)	0.9901 (0.2413)	0.9189 (0.2396)	0.8758 (0.2374)
Sharpe Ratio	2.869	1.934	3.896	3.627	3.478

Note: $\delta = 1, \beta = 0.01$. Standard deviation in parenthesis. Risk free rate 0.05

Table 2: Result Comparison

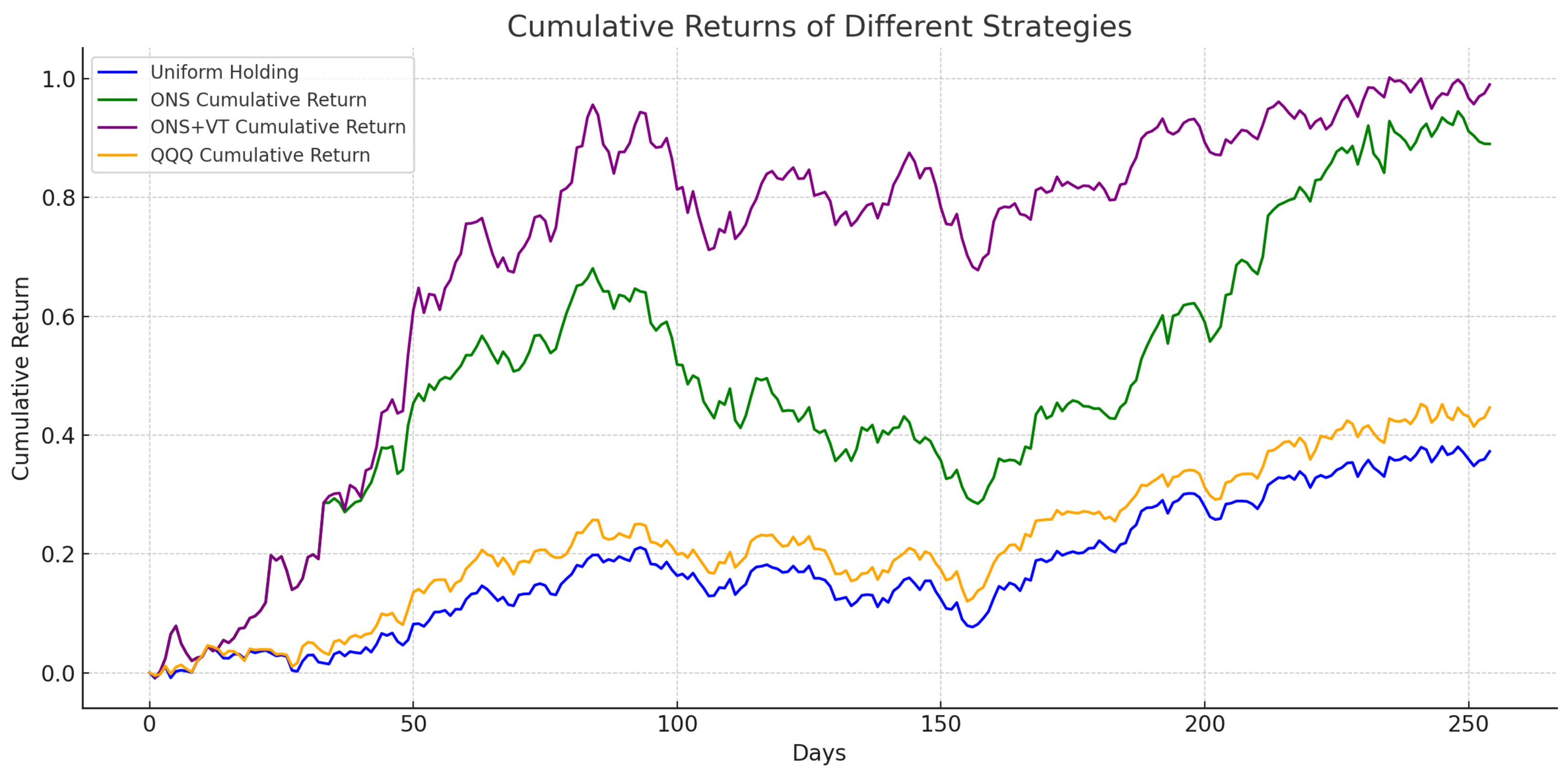

Figure 1 2023-2024 Portfolio Cumulative Return

- The cumulative return from ONS and ONS+VT were the same in the first 30 periods
- Using volatility threshold as side information, ONS+VT portfolio achieved higher return while sustain less drawdown
  - ➢ Robust risk management for no-shorting investment strategies

Reference

- Agarwal, A., Hazan, E., Kale, S., & Schapire, R. E. (2006). Algorithms for portfolio management based on the Newton Method. Proceedings of the 23rd International Conference on Machine Learning - ICML '06. https://doi.org/10.1145/1143844.1143846
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